



JVB Networking Labs

CDP & LLDP – Cisco Discovery Protocol and Link Local Discovery Protocol

CCNA Topic 2.3 - Configure and verify Layer 2 discovery protocols (CDP and LLDP)

Scenario

You are administering two running networks together through a single point-to-point link. All network devices are Cisco proprietary except for SW2. You wish to take advantage of layer 2 discovery protocols to obtain information about device neighbors.

Finally, all network devices should be configured to take advantage of a common layer 2 discovery protocol.

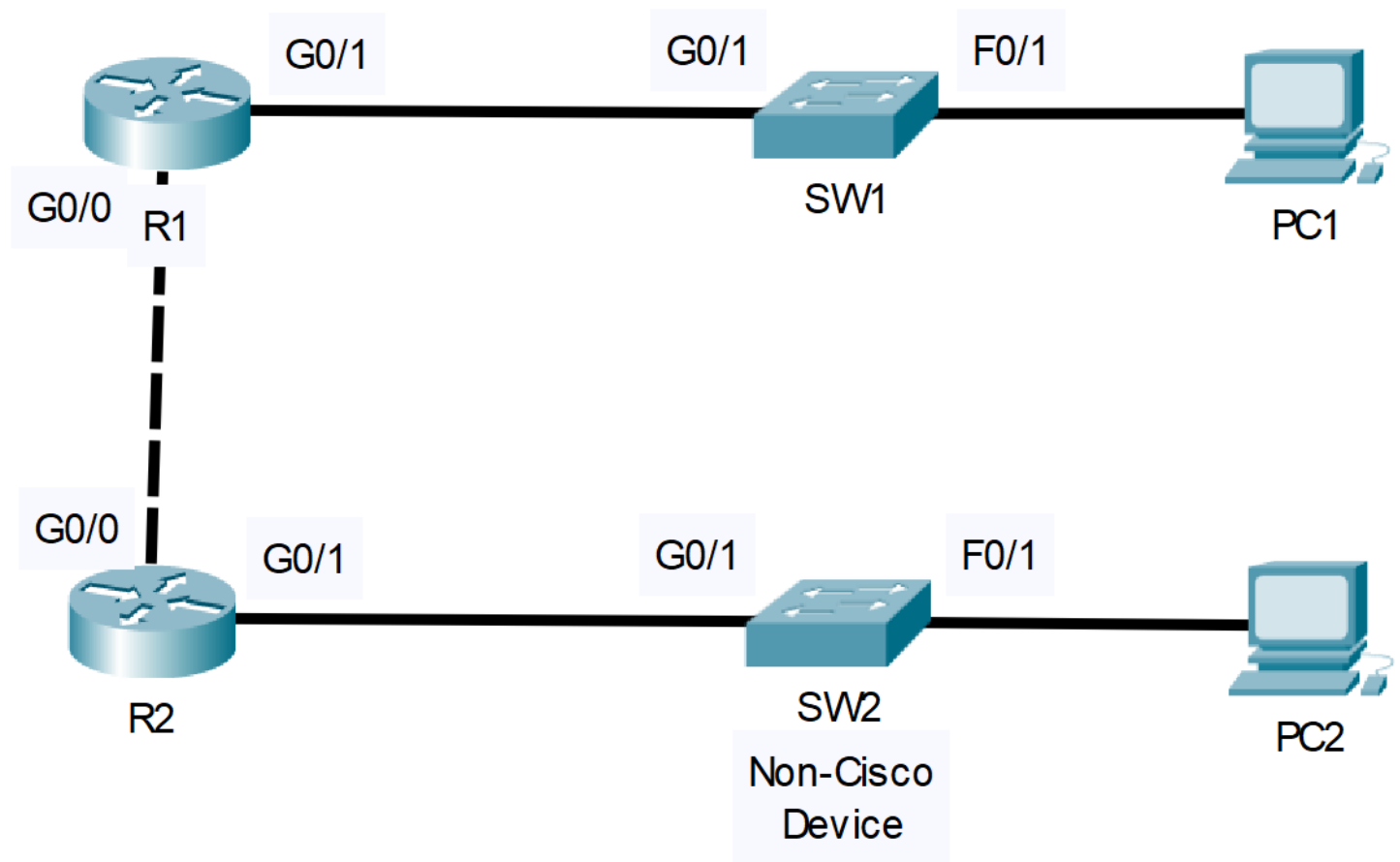
Requirements

- Demonstrate ability to view layer 2 discovery information via the CLI
- Disable CDP on all network devices that are using it, both globally and on the interface level
- Enable LLDP on all network devices, both globally and on the interface level

Post-Lab Testing

- Run appropriate 'show' commands to confirm exchange of layer 2 information

Network Topology



Devices

- 2 Routers, 2 Switches – All network devices Cisco proprietary except SW2

Configuration Plan Explanation

First, use the appropriate CLI commands to view discovery information on all network devices. Take note of outputs. *LLDP is usually disabled by default, so discovery commands on SW2 are not available until it is enabled later.*

Second, a common discovery protocol must be configured among the network devices. This means using LLDP, or Link Layer Discovery Protocol. CDP, or Cisco Discovery Protocol, is a Cisco proprietary protocol and will not share information with non-Cisco neighbor devices. CDP is enabled by default on most Cisco devices, so it must be disabled both globally and on the interface level on R1, R2, and SW1. All devices will then need to have LLDP enabled, globally and on the interface level.

Finally, run the appropriate CLI discovery commands once more on the network devices. Take note of the outputs and the differences.

DON'T FORGET TO SAVE YOUR CONFIGURATIONS!

Configuration

R1

enable

show cdp

show cdp traffic ***This command is often not available in network simulation software***

show cdp interface

show cdp neighbors

show cdp neighbors detail

configure terminal

no cdp run

interface range g0/0, g0/1

no cdp enable

lldp transmit

lldp receive

exit

lldp run

exit

show lldp

show lldp traffic ***This command is often not available in network simulation software***

show lldp interface ***This command is often not available in network simulation software***

show lldp neighbors

show lldp neighbors detail

write

SW1

enable

show cdp

show cdp traffic ***This command is often not available in network simulation software***

show cdp interface

show cdp neighbors

show cdp neighbors detail

configure terminal

no cdp run

interface range f0/1, g0/1

no cdp enable

lldp transmit

lldp receive

exit

lldp run

exit

show lldp

show lldp traffic ***This command is often not available in network simulation software***

show lldp interface ***This command is often not available in network simulation software***

show lldp neighbors

show lldp neighbors detail

write

R2

enable

show cdp

show cdp traffic ***This command is often not available in network simulation software***

show cdp interface

show cdp neighbors

show cdp neighbors detail

configure terminal

no cdp run

interface range g0/0, g0/1

no cdp enable

lldp transmit

lldp receive

exit

lldp run

exit

show lldp

show lldp traffic ***This command is often not available in network simulation software***

show lldp interface ***This command is often not available in network simulation software***

show lldp neighbors

show lldp neighbors detail

write

SW2

enable

configure terminal

lldp run

interface range f0/1, g0/1

lldp transmit

lldp receive

exit

exit

show lldp

show lldp traffic ***This command is often not available in network simulation software***

show lldp interface ***This command is often not available in network simulation software***

show lldp neighbors

show lldp neighbors detail

write

Verification Commands

Router

```
show lldp
show lldp traffic ***This command is often not available in network simulation software***
show lldp interface ***This command is often not available in network simulation software***
show lldp neighbors
show lldp neighbors detail
show running-config
```

Switches

```
show lldp
show lldp traffic ***This command is often not available in network simulation software***
show lldp interface ***This command is often not available in network simulation software***
show lldp neighbors
show lldp neighbors detail
show running-config
```

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More labs at



<https://github.com/JVanBurenIV/JVB-CCNA-Networking-Labs>



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